

The Binomial Distribution

Introduction

The binomial distribution is a discrete probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials. It is one of the most fundamental distributions in probability theory and is widely used in applications ranging from quality control to medical trials. Each trial has only two possible outcomes: success or failure.

Mathematical Definition

The probability mass function of a binomial distribution is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where n is the number of trials, k is the number of successes, p is the probability of success on each trial, and $\binom{n}{k}$ is the binomial coefficient “ n choose k ”.

Properties

The binomial distribution has several important characteristics. The mean is given by $\mu = np$, and the variance is $\sigma^2 = np(1 - p)$. The distribution is symmetric when $p = 0.5$ and becomes increasingly skewed as p moves away from 0.5.

The binomial distribution has numerous practical applications, including modeling the number of defective items in a sample, the number of correct answers on a multiple-choice test, and the number of heads in a series of coin flips. When n is large, the binomial distribution can be approximated by the normal distribution, which is useful for computational purposes.